Honghao Jia

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Education

Institute of Computing Technology, Chinese Academy of Sciences — Beijing, China
Jointly-Trained Master in State Key Lab of Processors; Research: AI Infra — *Sep. 2025 – Expected Jun. 2027*

University of Science and Technology of China — Hefei, China
Master of Microelectronics; GPA: 3.59 / 4.3 — *Sep. 2024 – Expected Jun. 2027*

Harbin Engineering University — Harbin, China
Bachelor of Computer Science and Technology; GPA: 89.33 / 100 — *Sep. 2020 – Jun. 2024*

Research Experience

AI Accelerators for Efficient Quantized Inference of LLMs | Research Project — *Sep. 2025 – Present*

- The research project is to design a adaptive delta quantization with hardware-software Co-Design for LLM inference, aiming to develop a novelty quantization hardware to balance accuracy and memory footprint reduction.
- Designed an EMA-based per-channel adaptation algorithm (ADKV) to track temporal drift (zero-point and scaling factor) in KV cache, while enabling delta coding to embed metadata into the quantized sequence, decoupling metadata overhead from context length and reducing metadata memory by $\sim$12.8$\times$.
- Developed an SGD-based calibration and fused ADKV-GEMV CUDA and Triton kernel, achieving accuracy comparable to FP16 with $\sim$1.96$\times$ inference throughput improvement.
- Implemented an ASIC accelerator based on systolic array (SA) architecture for ADKV, introducing ADKV encoder and decoder modules that are fully compatible with the SA dataflow without incurring additional latency.

Bergamot: A Superscalar RISC-V RV32GC Processor | Engineering Project — *Nov. 2023 – Jun. 2024*

- Implemented and validated Bergamot, a full-scale RISC-V RV32GC processor on Xilinx FPGA, capable of running the Linux kernel.
- Designed a multi-stage RISC-V processor with a 9-level pipeline using Chisel HDL, implementing frontend, execution, and backend pipelines, reducing HDL code size by $\sim$74% compared to Verilog and improving modularity and reusability.
- Developed a 2-way superscalar microarchitecture with an out-of-order execution pipeline and ROB-based commit, featuring 4 parallel functional units (ALU, FPU, Branch, and Memory), achieving up to 2 IPC peak throughput.
- Implemented key system components including Sv32 MMU (ITLB/DTLB), BTB-based branch prediction ($\sim$91% accuracy), and 2/4-way set associative caches (L1/L2, $\sim$87% hit rate), and an interrupt handling and MMIO framework, supporting the AXI4 bus protocol with integrated external peripherals (SPI, I2C, GPIO, . . .).
- Achieved 160+ stars on GitHub [github.com/LoveLonelyTime/Bergamot].

Internship Experience

Tencent Holdings Ltd. | WXG AI Infra Intern — *Mar. 2026 – Present*

- Implemented a distributed generative retrieval (GR) training and inference system, including the Qwen2.5 bi-encoder embedding model, distributed retrieval pipeline, and reranking module.
- Adopted inverted file index with product quantization (IVF-PQ) with asymmetric distance computation (ADC) on the GPU, while offloading retrieval to the CPU to design and optimize a GPU-CPU pipeline, achieving $\sim$2.4$\times$ speedup.
- Implemented data parallelism across GPUs by partitioning documents into parallel groups, where each GPU rank retrieves a subset of documents and the full loss is computed via distributed cross-entropy, achieving $\sim$4.6$\times$ speedup.
- Deployed the distributed GR system on NVIDIA H20 clusters (256 GPUs), enabling retrieval of 50 million documents (4 billion in total) within 0.7 s.

Cambricon Technologies Co., Ltd. | IC Backend Intern — *Sep. 2025 – Dec. 2025*

- Designed and optimized a 12 nm CMOS standard cell library focusing on area efficiency, using Cadence Virtuoso, Synopsys DC, ICC across the full custom digital design flow.

Software Research Institute, China Unicom Co., Ltd. | Software Engineering Intern — *Jun. 2022 – Sep. 2022*

- Built a user service order (eSIM) management system supporting order tracking, lifecycle management, and real-time status queries for internal operations.
- Developed a WeChat Mini Program frontend using WXML, JavaScript, and WXSS, delivering a lightweight and responsive mobile client.
- Redesigned and normalized database schema with MyBatis optimization, reducing 23% data redundancy and improving SQL performance by $\sim$3.4$\times$.
- Implemented a high-performance backend system with multi-level caching (Redis + MyBatis) and event-driven messaging (Kafka + MQTT), reducing latency by $\sim$4.2$\times$ and supporting CI/CD pipelines.

Publications

- **Honghao Jia**, Zhenxing Li, Jialiang Guo, Jiacheng Gan. ADKV: A Low-Overhead Adaptive Delta Quantization for KV Cache in LLM Inference. In *Advances in Neural Information Processing Systems*, 2026. (Under Review)

Honors and Awards

- **Bronze Award** in the ACM/ICPC (International Collegiate Programming Contest), Asia Regional Contest, 2022, 2023.
- **Gold Award** in the CCPC (China Collegiate Programming Contest), Heilongjiang Provincial Contest, 2023.
- **Silver Award** in the CCPC, Northeast Three Provinces Contest, 2023.
- **First-Class Academic Scholarship** at Harbin Engineering University, 2022.
- **First-Class Academic Scholarship** at the University of Science and Technology of China, 2024.

Research Interests

Computer Architecture; High Performance Computing; General Processor; AI Accelerator; Digital & Analog IC; FPGA; SoC; Embedded System; Signal Processing

Skills

- **Languages:** Chinese (Native), English (CET6: 512, TOEFL: 88, TOEIC: 790), Japanese (JLPT N1: 109)
- **Programming:** C/C++, Python, Java, Rust, JavaScript, Haskell, Verilog
- **Technologies:** AI Development (PyTorch, Transformers), GPU Programming (CUDA, Triton, shader programming), High-Performance Computing (MPI, NCCL, Ray, Megatron-LM), Hardware Design (FPGA, IC design), Programming Languages (Lambda calculus, Haskell), Web Development (JavaScript/TypeScript, HTML5, CSS, Vue, Spring)
- **Soft Skills:** Independent Research, Teamwork, Academic Writing